Development of Adhesive Strength, Heat Resistance, and Oil Resistance in Phenolic Resin Adhesives Modified with Copper Oxide Nanoparticles

Abstract

Phenolic resin (PF) adhesives are widely used in engineered wood composites and structural bonding applications due to their excellent heat resistance, chemical stability, and adhesive capabilities. However, the functional properties of PF adhesives can be further enhanced through nanomaterial modification. This paper reviews the development of adhesive strength, heat resistance, and oil resistance in PF adhesives modified with copper oxide (CuO) nanoparticles. The incorporation of nano-CuO into PF resin systems has been shown to reduce curing activation energy, accelerate addition and condensation reactions, and improve bonding performance in oriented strand board (OSB) and plywood applications. A 40% $CuCl_2$ concentration-derived nano-CuO gave a 74% increase in perpendicular MOR of OSB. However, the effects on water resistance and oil resistance require further optimization, as polyvinyl alcohol (PVA) incorporation has been shown to reduce water resistance of cured PF resins. The simultaneous in-situ preparation strategy offers a cost-effective route to functional PF adhesives with potential bio-resistance and improved mechanical properties.

Keywords: Phenolic resin; Copper oxide nanoparticles; Adhesive strength; Heat resistance; Curing kinetics; Oil resistance

1. Introduction

Phenol-formaldehyde (PF) resin was the first thermosetting resin synthesized in 1907 and has since been shown to possess many good properties including resistance to heat, corrosion, wear, and excellent adhesive-related capabilities . In engineered wood composites, PF adhesives serve as the critical binding phase that determines the mechanical integrity, durability, and functional performance of the final product.

The modification of PF resins with nanomaterials represents a promising approach to enhance their functional properties beyond the baseline performance. Nano-copper oxide (CuO) has attracted particular attention due to its ability to reduce the curing activation energy of PF resin, improve curing degree, and impart bio-resistance to wood composites . The economic barrier of nano-CuO powder—which is significantly more expensive than copper salts—has been addressed through the development of simultaneous in-situ preparation strategies, where nano-CuO is generated during PF resin synthesis using inexpensive copper salt precursors .

This paper reviews the development of adhesive strength, heat resistance, and oil resistance in nano-CuO-modified PF adhesives, drawing on the extensive body of work by Gao, Du, Yi, Zhao, Guo, and colleagues who have systematically studied this system.

2. Adhesive Strength Development

2.1 Curing Kinetics Enhancement

The adhesive strength of PF resins is fundamentally governed by the curing process—the crosslinking reactions that transform the resin from a thermoplastic to a thermosetting network. The introduction of nano-CuO has been shown to exert a positive effect on the curing reaction of PF resin, characterized by a significant decrease in the activation energy of the reaction .

Dynamic DSC analysis reveals that the curing exothermic peaks of PF resins are concentrated in the 300–340°C range . Loading levels of 4% and 8% derived nano-CuO reduced the apparent activation energy of the PF resin and accelerated both the addition and condensation reactions . The reduction in activation energy was most significant within a conversion degree of 65% .

The mechanism underlying this acceleration involves a complexation reaction between copper ions and oxygen on the phenolic hydroxyl groups, which influences phenoxy carbons as shown by ^{13}C NMR . Quantitative analysis indicated that nano-CuO had a positive effect on para-substituted reactions, thereby enhancing both addition and condensation reactions .

2.2 Bonding Performance in Wood Composites

The practical manifestation of improved curing kinetics is enhanced bonding strength in wood composites. A study examining the effect of cupric chloride concentration on resin performance found that nano-CuO derived from a 40% concentration of $CuCl_2$ gave a **74% increase in perpendicular MOR** of oriented strand board (OSB) . This represents a substantial improvement in the load-bearing capacity of the bonded composite.

The water resistance and bonding strength of glued OSB prepared with nano-CuO-modified PF resins indicated that the influence of nano-CuO on the mechanical properties of the boards was positive . However, the shear strength under several test conditions decreased slightly, though values remained within the requirements set by the appropriate Chinese Standard .

2.3 Adhesive Strength in Different Composite Systems

In flakeboard applications, nanosize copper oxide had no or slightly beneficial effects on the mechanical properties and dimensional stability of the boards . The DSC results indicated that the comparable activation energy and reaction enthalpy obtained from nanosize copper oxide treated samples contributed to the higher degree of conversion of PF resin and undiminished bonding strength of flakeboards . This was attributed to the acceleration of addition reactions among small molecules in PF resin and the enhancement of heat transfer of the composite mat caused by nanosize copper oxide .

3. Heat Resistance Development

3.1 Thermal Degradation Behavior

Thermogravimetric (TG) analysis has been used to evaluate the thermal stability of nano-CuO-modified PF resins. The main degradation peak of the modified PF resins **narrowed and shifted to a lower temperature** by introducing nano-CuO . This shift suggests that the presence of nano-CuO alters the thermal degradation pathway of the resin, potentially promoting volatilization of degradation products or modifying the crosslinked structure.

The narrowing of the degradation peak indicates a more homogeneous degradation process, which may be related to the more uniform crosslinked network achieved through enhanced curing. While the shift to lower temperature might initially appear detrimental to heat resistance, it likely reflects improved heat transfer within the composite due to the thermally conductive CuO nanoparticles .

3.2 Dynamic Mechanical Thermal Analysis

Dynamic mechanical thermal analysis (DMTA) of aqueous PF resin modified by nano-CuO containing alkane surfactant and polyvinyl alcohol (PVA) 17-99 revealed that the addition of nano-CuO incorporating PVA separated the gel point and vitrification point in the tan delta curve . This separation is related to the delaying of moisture loss in modified PF resin during the curing process .

The modification showed minimal influence on modulus of rupture (MOR) and modulus of elasticity (MOE), indicating that the heat resistance enhancement did not come at the expense of mechanical performance . The DMTA results suggest that nano-CuO modification provides a pathway to tailor the curing and thermal behavior of PF resins for specific application requirements.

4. Oil Resistance and Water Resistance Considerations

4.1 Water Resistance Challenges

Water resistance is a critical performance parameter for PF adhesives in exterior applications. The incorporation of nano-CuO with PVA 17-99 was found to **reduce the water resistance** of cured PF resins . This adverse effect is attributed to the hydrophilic nature of PVA, which was used as a dispersing agent for the nano-CuO.

However, under specific test conditions—dipping in 100°C water for 6 hours, drying for 20 hours at 63°C in air, followed by dipping in 100°C water for 4 hours—PVA consolidated the re-curing effect on the PF resin and compensated for the strength loss from hydrolysis . This

suggests that the modified PF system has the potential for developing self-curing wood composites for structural construction applications .

4.2 Implications for Oil Resistance

While direct studies on oil resistance of nano-CuO-modified PF adhesives are limited, several factors suggest potential benefits. The enhanced crosslinking density achieved through accelerated curing would generally improve resistance to oil penetration. Additionally, the presence of nano-CuO particles may create tortuous pathways that impede oil diffusion through the resin matrix.

The complexation between copper ions and phenolic hydroxyl groups may also contribute to oil resistance by reducing the availability of polar sites for oil interaction. However, systematic studies specifically evaluating oil resistance are needed to establish quantitative performance benchmarks.

5. Simultaneous In-Situ Preparation Strategy

5.1 Process Innovation

The cost of nano-CuO powder is significantly higher than copper salt precursors, creating an economic barrier to widespread adoption. To address this, Yi et al. developed a strategy where PF resin synthesis and nano-CuO preparation are carried out **simultaneously in one system** . Nano-CuO was prepared by a direct precipitation method, with copper salt solution added into the basic alkaline reactor during PF resin synthesis .

5.2 Morphological and Structural Characteristics

XPS results proved that nano-CuO was present in the modified PF resin . XRD data showed that the derived nano-CuO has a **face-centered cubic structure** . The minimum particle size of

nano-CuO was **below 10 nm**, and the distribution of CuO was uniform as seen in TEM and EDXS images .

Four different types of nano-CuO crystal structures were observed, varied according to the copper salts used ($CuBr_2$, $CuCl_2·2H_2O$, $Cu(NO_3)_2·3H_2O$, and $CuSO_4·5H_2O$) . This morphological diversity provides opportunities for tailoring the functional properties of the modified PF resins.

5.3 Performance Outcomes

The simultaneous in-situ preparation strategy has been validated through comprehensive performance evaluation. The modified PF resins were evaluated via XPS, XRD, TEM, EDXS, and TG, with physical and mechanical properties of OSB also assessed . Though the physical and mechanical properties of OSB decreased slightly, values were still within the requirements set by the appropriate Chinese Standard, meaning that the modified PF resin has potential to be applied in engineered wood composites manufacturing .

6. Summary and Perspectives

6.1 Key Findings

The modification of PF adhesives with nano-CuO offers multiple performance benefits:

1. **Adhesive Strength**: Nano-CuO reduces curing activation energy and accelerates both addition and condensation reactions. A 74% increase in perpendicular MOR of OSB has been demonstrated with optimized $CuCl_2$ concentration .

2. **Heat Resistance**: The main degradation peak of modified PF resins narrows and shifts to lower temperature, indicating altered thermal degradation behavior with improved heat transfer characteristics .

3. **Water Resistance**: The incorporation of PVA as a dispersing agent reduces water resistance, but re-curing effects can compensate for strength loss under specific conditions .

4. **Cost Efficiency**: Simultaneous in-situ preparation using inexpensive copper salts reduces the cost barrier for nano-CuO-modified PF resin production .

6.2 Future Research Directions

Several areas warrant further investigation:

1. **Oil Resistance**: Systematic studies on oil resistance of nano-CuO-modified PF adhesives are needed to establish performance benchmarks.

2. **Dispersion Optimization**: The adverse effect of PVA on water resistance suggests that alternative dispersing strategies should be explored.

3. **Long-Term Durability**: Extended exposure testing is needed to validate the long-term performance of modified PF adhesives in exterior and industrial applications.

4. **Scale-Up Feasibility**: The industrial feasibility of the simultaneous in-situ preparation process requires verification through pilot-scale trials.

References

[1] S. Zhao, W. Gao, T. Yi, C. Guo, K. Zhan, G. Du, Fabrication of nano-cupric oxide in phenol–formaldehyde resin adhesive: synthesis, curing characteristics, and morphology, Wood Sci. Technol. 53 (2019) 819–835.

[2] W. Gao, J.J. Morrell, S. Zhao, H. Lei, T. Yi, S. Xia, Q. Lu, K. Zhan, C. Guo, Fabrication of nano-cupric oxide in phenol–formaldehyde resin adhesive: effect of cupric chloride concentration on resin performance, Wood Sci. Technol. 54 (2020) 1551–1567.

[3] T. Yi, S. Zhao, W. Gao, C. Guo, L. Yang, G. Du, The similar in-situ polymerization of nano cupric oxide preparation and phenol formaldehyde resin synthesis: The process and mechanism, Int. J. Adhes. Adhes. 87 (2018) 109–118.

[4] T. Yi, C. Guo, S. Zhao, K. Zhan, W. Gao, L. Yang, G. Du, The simultaneous preparation of nano cupric oxide (CuO) and phenol formaldehyde (PF) resin in one system: aimed to apply as wood adhesives, Eur. J. Wood Wood Prod. 78 (2020) 471–482.

[5] W. Gao, C. Guo, T. Yi, S. Zhao, G. Du, Dynamic mechanical thermal analysis (DMTA) of aqueous phenol formaldehyde (PF) resin modified by nano copper oxide (CuO), Eur. J. Wood Wood Prod. 76 (2018) 1145–1151.

[6] C. Guo, T. Yi, J.J. Morrell, et al., Incorporation of a nano/micro CuO formulation into phenol formaldehyde (PF) resin: Curing kinetics, morphological analysis, and application, J. Wood Chem. Technol. 39 (2019) 372–383.

[7] W. Gao, J. Cao, D.P. Kamdem, Effect and mechanism of nanosize copper oxide on some physical and mechanical properties of flakeboards, Maderas Cienc. Tecnol. 13 (2011) 203–210.

[8] W. Gao, G. Du, Dynamic mechanical analysis (DMA) of nano cupric oxide (CuO) modified aqueous phenol formaldehyde (PF) resin, IRG/WP 14-40654, 2014.